Harshit Singh

Data Analyst | SQL • Python • Statistical Modeling • Time-Series Forecasting

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Education

Vellore Institute of Technology, Chennai — Chennai, India
B.Tech, Electronics and Computer Engineering — CGPA: ***7.97*** — *2023 – 2027*

New Era School (CBSE) — Ghaziabad, Uttar Pradesh
Class XII: ***87.16%*** *— Class X:* ***89.83%*** — *2020 – 2023*

Experience

AI & Robotics Intern — Autonomous Mobile Robotics [**Certificate**] [**GitHub**] — Jul 2025 – Aug 2025
STEMTEC AI & Robotics Technology Pvt. Ltd. — *Chennai, India*

- Benchmarked positional and orientation drift by running time-series error analysis on **15 Hz** Extended Kalman Filter telemetry against raw encoder logs, quantifying a **60% reduction** in mean spatial error (**0.45 m → 0.18 m**) and **69.2%** in angular error across 10 m ground-truth trials.
- Analyzed high-frequency LiDAR telemetry (∼**3,600 points/sec** at **10 Hz**) to validate a windowed sector-filtering rule, achieving **95% obstacle-detection accuracy** across 20 randomized stress-test trials with **sub-50 ms** end-to-end telemetry latency.

Projects

TRIDENT — Risk & Spatial Emergency Analytics *(Python, NetworkX, SQLite)* [**Demo**] [**GitHub**] [**Live**]

- Auto-prioritized emergency requests into **5 severity tiers** using a Multi-Criteria Decision Analysis (MCDA) scoring model weighing injury counts, vulnerable demographics, and event type.
- Forecasted 24-hour hazard trends (**MSE < 0.005**) from **17,500+** hourly records via an automated ETL pipeline, and flagged municipal drainage breaches by modeling peak network throughput.

NeuroBridge AI — Clinical Analytics Platform *(SQL, Node.js)* [**GitHub**] [**Live**]

- Segmented patient populations into risk tiers and benchmarked therapy outcomes via multi-table SQL aggregation queries across **15+ relational tables**, matching patients to cohorts within ±**15** risk-score bands.
- Forecasted 30-day patient risk trajectories via least-squares linear regression over rolling 5-session windows, auto-triggering clinical alerts when risk slopes exceeded **+5 units/session**.

GNOME Stocks Suite — Financial Data Engine *(Python, Flask, SQLite)* [**GitHub**] [**Live**]

- Cleaned and indexed **150,000+** raw market instruments down to **2,500+** high-liquidity NSE equities, cutting ticker lookup latency by **99%** (**800ms → <5ms**).
- Standardized inconsistent symbol formats across **3** disparate data providers (Yahoo Finance, Finnhub, Angel One) into a unified schema, achieving **100%** metadata consistency for downstream reporting.

SPARC — Sign Language Recognition System *(OpenCV, Deep Learning)* [**Demo**] [**Web-App**]

- Reduced input dimensionality by **99.97%** (**921,600 → 229** engineered geometric features — joint angles + pairwise distances), enabling real-time inference at **92.4% accuracy**, 15–20 FPS.
- Cut false-positive classification errors by **92%** across **41** gesture categories via temporal sliding-window filtering — recognized as a **Top 5 national finalist** (500+ teams) at Smart India Hackathon 2025. [**Certificate**]

Skills

Languages: Python, SQL, C, Java
Data & Analytics: Pandas, NumPy, Scikit-Learn, Time-Series Forecasting, Statistical Modeling, Feature Engineering
Databases & Tools: SQLite, Query Optimization, ETL Pipelines, REST APIs, Git, Excel
Visualization: Chart.js, Recharts, Dashboarding
Professional Skills: Data-Driven Problem Solving, Cross-Functional Collaboration, Communication, Time Management

Achievements & Leadership

- **3rd Place, Technovation 2025** (Kurukshetra '25, CEG Guindy, Anna University) — ranked **2nd Runner Up out of 50+** teams; **INR 15,000** award. [**Certificate**]
- **Winner, Innovate X Impact (V-NEST)** — secured **1st Position out of 35** teams; **INR 5,000** award.
- **Deputy Operations Lead, Team Namo Nirvana (VIT Chennai)** — led **100+ member** teams across **5–6 competitions/semester**, boosting efficiency by **35%** and securing **2 sponsors.** [**LOA**] [**Certificate**] [**LOR**]

Languages

English (Fluent) | **Hindi** (Native)